Gopal Sharma

Senior Data Scientist • RAG & GraphRAG • LLM Systems • M.Sc. Physics

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Summary

Senior Data Scientist with 4+ years of applied ML, the last 3 shipping production Generative AI at Newgen Software. I build retrieval and reasoning systems and measure them: a single-first query router lifted multi-hop QA from 70.2% to 76.9% on 952 questions; a bidirectional graph walk raised deep multi-hop accuracy from 47% to 53% while cutting candidate paths 72%; and an async inference platform serves 7 NLP models at ~8,400 emails/hour per node. Physics-trained (CSIR-UGC NET AIR 81), comfortable from first-principles modelling to production serving.

Experience

Senior Data Scientist Oct 2025 – Present

Newgen Software Technologies Ltd., New Delhi

- **Multi-hop GraphRAG.** Designed a bidirectional meet-in-the-middle search over a 52,474-node / 60,475-edge knowledge graph that answers multi-hop questions without a known target entity; raised accuracy from 47% to 53% (70-question clean set) and correct-path selection from 39% to 72–84%.
- Built a fully local answer-candidate generator (reachable-set BFS, embedding filter, cross-encoder rerank), cutting candidate paths 72% (~1,500 → ~420) with zero API calls.
- **Query routing.** Showed query decomposition is a rescue, not a default (always-decompose fell 70.2%→62.8%); a single-first router that decomposes only on abstention reached 76.9%, invoking the 3×-cost path on just 18% of queries.
- **Regulatory knowledge graph.** Built a resumable, layered graph over 371 financial-regulation documents (17k pages): 46,508 entities / 109,572 relations for $35.1 ($2.03 per 1,000 pages); doc recall@20 0.93. A fully on-prem rebuild (Qwen3 on vLLM + bge-m3) matched the cloud-built graph (doc hit@20 0.91 vs 0.86).
- Migrated the graph backend from NetworkX to Neo4j via a ~150-line adapter, reproducing 25/25 query results exactly and loading the full graph in 7 s.
- **Document AI.** Built a CPU-only, LLM-free layout-aware PDF chunker (DocLayNet YOLO, typography-based heading levels, OCR fallback): page hit@5 of 85.5% on OHR-Bench (vs 95.5% with ground-truth text). Table structure recognition (Table Transformer + token-driven post-processing) reached row boundary-F1 0.72 and column IoU-F1 0.94 on financial statements.
- **Platforms.** Architected an async email-intelligence platform (FastAPI, Celery, Redis) running 7 NLP models (toxicity, PII, complaint type, sentiment); a multiprocessing engine keeps 28 transformers preloaded (~5 GB), bypassing the GIL; k6 load test: 1.29 s median latency at 16 concurrent users, 58% CPU.
- Split a monolithic ingestion pipeline into 6 async microservices (acquisition, OCR, chunking, cleaning, embedding, storage) over 4 enterprise sources and 6 formats, with hybrid Milvus search for air-gapped, 100% on-prem clients. Shipped tender technical-evaluation and similar-tender search APIs with page-level evidence.
- Mentor junior data scientists; partner with product and backend teams to take GenAI from prototype to production.

Data Scientist Oct 2023 – Oct 2025

Newgen Software Technologies Ltd., New Delhi

- Implemented LLM guardrails with 25+ parallel input, retrieval and output scanners (1–2 s): prompt injection, jailbreaks, 27 secret types, hidden Unicode, PII (incl. Aadhaar/PAN), toxicity, bias, factual consistency and hallucination across 22 languages.
- Ran Llama-3 teacher→student distillation for text-to-SQL on 1,500 queries (4-bit student 54.6% vs teacher 83.3%) and assessed 8+ compression methods (QLoRA, AWQ, GPTQ, PEQA, LLM-QAT, pruning, low-rank).
- Standardized LLM evaluation on 7 metrics (SQuAD EM/F1, SAS, LERC, QAFactEval, QuestEval, BBH, MMLU) chosen by ground-truth availability; generated synthetic training data for 16 NLP tasks from 7 source formats.

Data Scientist / AI-ML Solution Analyst Jul 2022 – Aug 2023

Number Theory (acquired by Newgen Software), Gurugram

- Built a 143-function feature library for a loan-risk scorecard (5 credit products, 100,000 applicants, 4 sources: bank statements, tax returns, bureau; 15-day to 12-month windows).
- Trained PySpark GBT and Random Forest models per product, tuning thresholds for maximum default recall at 0.70 precision; modelled customer value and propensity for loans and credit cards.

Open-Source Projects

- **openai-budget-guard** — per-key, per-task token and cost budgets for OpenAI keys: atomic preflight reservations (concurrent calls cannot overshoot), streaming-usage metering, SQLite ledger, drop-in proxy and CLI; openai SDK 1.x–3.x.
- **graphwalk-rag** — graph RAG that walks: provenance-preserving LightRAG build, NetworkX/Neo4j, bidirectional meet-in-the-middle traversal. Gold-path recall 100% vs 88.6% for enumeration with ~200× fewer candidates (70 multi-hop questions, 52k-node graph); randomized tests prove equivalence to brute force.
- **layout-rag** — CPU-only, LLM-free layout-aware PDF chunking (typography-derived heading levels, header-carrying table rows, budgets in embedder tokens) with hybrid dense+BM25 retrieval and local reranking; benchmark harness vs LangChain and Docling chunkers on OHR-Bench / FinanceBench.
- **qft-renorm-ai** — my M.Sc. QED renormalization in code: SymPy derives Z-factors, the Ward identity $Z_1 = Z_2$ and $\beta(e) = e^3/12\pi^2$ in $n = 4 - \epsilon$. Sparse regression on the RG equation rediscovers $\beta_0 = 11 - 2n_f/3$ from noisy $\alpha_s(Q)$ data (asymptotic freedom lost at $n_f \approx 16.3$ vs 16.5); CAS tools served over MCP.
- **portfolio** — live knowledge-graph demo: visitors query a Qwen3-14B-built graph of my work (892 entities) and watch a meet-in-the-middle search return the top-5 evidence paths in about 4 ms, fully in the browser (transformers.js embeddings, no backend).
- **rag-mcp-server** — document search and knowledge-graph walks as Model Context Protocol tools for any AI assistant (stdio and HTTP).

Education

M.Sc. Physics, National Institute of Technology Calicut 2017 – 2019
CGPA 8.74/10. Project: *Renormalization of Quantum Field Theory* — QED Feynman rules, tree-level S-matrix (pair production, Compton, Møller, Bhabha), Ward–Takahashi identity, one-loop self-energies in dimensional regularization.

Diploma, Artificial Intelligence & Machine Learning, University of Hyderabad 2021 – 2022
B.Sc. Physical Science, S.G.T.B. Khalsa College, University of Delhi (82.3%) 2014 – 2017

Skills

GenAI & LLMs: RAG, GraphRAG, multi-hop QA, query decomposition & routing, LLM evaluation, guardrails, fine-tuning, distillation, quantization (QLoRA, AWQ, GPTQ), synthetic data
ML & NLP: PyTorch, Hugging Face, sentence-transformers, cross-encoder reranking, NER, classification, DETR/Table Transformer, YOLO, OCR (Tesseract, ABBYY), PySpark MLlib, scikit-learn
Retrieval & Data: LightRAG, Neo4j, NetworkX, Milvus, hybrid search, PostgreSQL, MongoDB, Redis
Engineering: Python, FastAPI, Celery, asyncio, multiprocessing, Docker, vLLM, k6, Git, Linux, C++

Achievements

CSIR-UGC NET (Physical Sciences) All India Rank 81, 2019 • IIT JAM (Physics) All India Rank 469, 2017